Algorithm Sample 1

Dataset-expansion sample

1 Procedure

The pseudocode below is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

1: For $\beta \in (1, \frac{3}{2})$ set $\varepsilon = \frac{3}{2} - \beta$ and $T = L^{2(1-\varepsilon)}$. For $i = 1, \cdots, d$, solve for the approximate first-order corrector $\phi_{i,T}^{(L)}$:

$$\frac{1}{T}\phi_{i,T}^{(L)} - \nabla \cdot a\nabla\phi_{i,T}^{(L)} = \nabla \cdot ae_i \text{ in } Q_{2L}, \qquad \phi_{i,T}^{(L)} = 0 \text{ on } \partial Q_{2L}.$$

2: Calculate the approximate homogenized coefficients via

$$a_h^{(L)}e_i = \int \omega q_{i,T}^{(L)},$$

where

$$q_{i,T}^{(L)} := a(e_i + \nabla\phi_{i,T}^{(L)})$$

and $\omega(x) = \frac{1}{L^d}\hat{\omega}(\frac{x}{L})$ with $\hat{\omega}$ as in Theorem **??**.

3: Find $\tilde{u}_h^{(L)}$ on ∂Q_L:

$$\tilde{u}_h^{(L)} = \int G_h^{(L)} * (\nabla \cdot g),$$

where $G_h^{(L)}(x) := \frac{1}{4\pi\left|(a_h^{(L)})^{-1/2}x\right|}$ is the Green function for the constant-coefficient operator $-\nabla \cdot a_h^{(L)}\nabla$.

4: Solve for approximate first-order flux correctors $\sigma_{i,T}^{(L)} = \{\sigma_{ijk,T}^{(L)}\}_{j,k}$:

$$\frac{1}{T}\sigma_{ijk,T}^{(L)} - \Delta\sigma_{ijk,T}^{(L)} = \partial_j q_{ik,T}^{(L)} - \partial_k q_{ij,T}^{(L)} \text{ in } Q_{\frac{7}{4}L}, \qquad \sigma_{ijk,T}^{(L)} = 0 \text{ on } \partial Q_{\frac{7}{4}L}.$$

5: Solve for approximate second-order correctors $\psi_{ij,T}^{(L)}$:

$$\frac{1}{T}\psi_{ij,T}^{(L)} - \nabla \cdot a\nabla\psi_{ij,T}^{(L)} = \nabla \cdot (\phi_{i,T}^{(L)}a - \sigma_{i,T}^{(L)})e_j \text{ in } Q_{\frac{3}{2}L}, \qquad \psi_{ij,T}^{(L)} = 0 \text{ on } \partial Q_{\frac{3}{2}L}.$$

6: For the indices

$$(i,j) \in \mathcal{J} = \{(1,2),(1,3),(2,3),(2,2),(3,3)\},$$

calculate

$$c_{ij,T}^{(L)} = -\int g \cdot \nabla\Big(\sum_{k=1}^{3}\phi_{k,T}^{(L)}\partial_k v_{h,ij}^{(L)} + (2-\delta_{ij})(\psi_{ij,T}^{(L)} - \frac{a_{hij}^{(L)}}{a_{h11}^{(L)}}\psi_{11,T}^{(L)})\Big),$$

where $v_{h,ij}^{(L)}$ denote the $a_h^{(L)}$-harmonic polynomials

$$v_{h,ij}^{(L)} = (1 - \frac{1}{2}\delta_{ij})(x_i x_j - \frac{a_{hij}^{(L)}}{a_{h11}^{(L)}} x_1^2).$$

7: Obtain $u_h^{(L)}$ as

$$u_h^{(L)} = \tilde{u}_h^{(L)} + \sum_{i=1}^{3} (\int g \cdot \nabla \phi_{i,T}^{(L)}) \partial_i G_h^{(L)} + \sum_{(i,j) \in \mathcal{J}} c_{ij,T}^{(L)} \partial_{ij} G_h^{(L)}.$$

8: Solve for $u^{(L)}$ (here and for the rest of the paper we adopt Einstein's summation convention for repeated indices):

$$-\nabla \cdot a \nabla u^{(L)} = \nabla \cdot g \text{ in } Q_L, \qquad u^{(L)} = (1 + \phi_{i,T}^{(L)} \partial_i + \psi_{ij,T}^{(L)} \partial_{ij}) u_h^{(L)} \text{ on } \partial Q_L.$$